



Boundary-Spanning Knowledge Search and Absorptive Capacity in Cooperative Innovation: A Study on Non-Core Firms in the Context of Sustainable Development

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Received: 15 January 2024 / Accepted: 22 March 2024 / Published online: 1 April 2024

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Abstract

This research paper investigates the vital role of boundary-spanning knowledge search and absorptive capacity in shaping the cooperative innovation performance of non-core firms operating in the context of sustainable development. The dynamic nature of technological innovation, exacerbated by market fluctuation and the challenges posed by the COVID-19 pandemic, has compelled small and medium-sized enterprises (SMEs) to seek innovative breakthroughs. Cooperative innovation emerges as a strategic avenue for SMEs to mitigate risks and enhance their willingness to innovate. However, non-core firms often encounter technological and market disadvantages, hindering innovation efforts. This study explores how boundary-spanning knowledge search, which bridges the gap between non-core firms and the external environment, influences cooperative innovation. Furthermore, the research delves into the mediating role of both potential and actual absorptive capacity in the relationship between boundary-spanning knowledge search and cooperative innovation performance. Empirical findings indicate that boundary-spanning knowledge searches positively impact cooperative innovation, with technological and market knowledge searches yield significant results. Moreover, absorptive capacity exhibits a dual mechanism, directly influencing cooperative innovation and mediating the relationship between knowledge search and performance. This paper extends existing research by focusing on non-core firms within the sustainable development context, offering valuable insights for SMEs seeking innovative strategies. It also highlights the diverse pathways through which boundary-spanning knowledge search affects cooperative innovation. These findings offer practical implications for enhancing non-core firms' innovation capabilities, including encouraging proactive knowledge search and developing absorptive capacity-inducing mechanisms through employee training and information platforms.

Keywords Cooperative innovation · Non-core firms · Boundary-spanning knowledge search · Absorptive capacity · Sustainable development

JEL Classification L25 · M11 · O32

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Introduction

The rapid iteration of industrial technology cycles has become a distinctive feature in the innovation environment and is influenced by sustainable development and market flattening trends (Zhang et al., 2019; Humphreys and Imas 2022). COVID-19 reveals the uncertainty of global economic development and has severely impacted the survival and development of most SMEs (Afonso, 2023; Wang et al., 2023a). Non-core firms are defined as SMEs at the edge of the innovation network; they have significant latecomer characteristics due to objective factors such as firm size and technological capabilities and subjective factors such as market discrimination and consumer stereotypes (Zang et al., 2022). Non-core firms face technological and market disadvantages, especially in a sustainable development context powered increasingly by bottom-up efforts. Deconstructing and absorbing heterogeneous resources becomes a key path for them to break the top-down innovation framework, achieve innovation breakthroughs, and catch-up technologically (Humphreys & Luk, 2022; Zang et al., 2022). Sustainable development requires non-core firms to improve resource efficiency and further achieve industrial upgrading and digital transformation (Su & Wu, 2024; Wen & Zhang, 2024). Cooperative innovation can help non-core firms meet these requirements, as it is a relatively convenient approach to coordinate specialized behaviors of different actors and enhance innovation value (Hindi & Frenkel, 2022; Malherbe, 2022).

However, even after joining a cooperative innovation network, the actual performances of non-core firms remain mixed (Kapoor & Agarwal, 2017; Sendawula et al., 2023). Some non-core firms have mitigated their disadvantages by making key decisions, such as iterating core technology capabilities and repositioning themselves in the market (Zang et al., 2022). However, many non-core firms are perceived as affiliated with core firms (or their subsidiaries) or being “foundry” or OEM enterprises and are trapped in a “low-end lock-in” mentality (Peng & Liu, 2021). This highlights the importance of identifying the actions that can address the dilemmas faced by non-core firms and effectively improve the quality of their cooperative innovation.

Many factors influence cooperative innovation performance; however, knowledge resources are consistently an essential element (Quintana-Garcia & Benavides-Velasco, 2008; Subramaniam & Loganathan, 2022). For non-core firms in the context of sustainable development, the ability to engage in boundary-spanning knowledge search provides an important bridge to cross organizational boundaries and broaden an enterprise’s systematic knowledge framework (Leone et al., 2022; Nerkar & Miceli, 2016). It also provides a clear goal orientation and is an important way for non-core firms to acquire heterogeneous knowledge (Lichtenthaler & Lichtenthaler, 2009). Previous studies have applied different perspectives to explore the relationship between boundary-spanning knowledge search and cooperative innovation performance.

For example, with respect to the knowledge spillover effect, firms conduct boundary-spanning knowledge searches to capture the current state of industry technology development (Tan et al., 2017). This helps clarify the direction of innovation breakthroughs and cooperative innovation goals. With respect to resource

dependence theory, firms obtain real-time market information through boundary-spanning knowledge search, helping them respond to market demand and reduce the risk of cooperative innovation failure (Zhang et al., 2016). Many previous studies applying these theories have examined pan-type enterprises, classifying the theoretical dimensions using search distance and depth criteria (Lin & Li, 2013; Rosenkopf & Nerkar, 2001). However, when studies focus on a specific target group of non-core firms, the research does not reflect the more universal characteristics of existing research findings due to their network status and lack of resource endowment (Zang et al., 2022).

Absorption of heterogeneous knowledge resources is a key aspect of cooperative innovation. Boundary-spanning knowledge search emphasizes the innovation connection that occurs between non-core firms and the external environment (Nerkar & Miceli, 2016; Zang et al., 2022). In contrast, absorptive capacity focuses on a series of processes within non-core firms, such as digesting and transforming heterogeneous knowledge into cooperative innovation results (Cohen & Levinthal, 1990). Absorptive capacity is an important channel for effectively identifying and utilizing knowledge resources, and it can optimize collaboration patterns among cooperative innovation agents and increase the efficiency of innovation resource transformation (Santoro et al., 2020). Previous studies confirmed that absorptive capacity significantly impacts the ability to take in external heterogeneous resources and is an important medium for transforming knowledge resources into innovation performance (Atuahene-Gima, 2003; Duan et al., 2021). Absorptive capacity can enhance the efficiency of boundary-spanning knowledge search by effectively identifying the available value of high-quality knowledge resources and integrating and applying them to form new products and services (Fosfuri & Tribo, 2008). Given the specific influencing mechanism of absorptive capacity on knowledge-based resources, it is mutually reinforcing to enhance the strength of absorptive capacity and increase the efficiency of boundary-spanning knowledge search (Schweisfurth & Raasch, 2018). As such, absorptive capacity may play a mediating role between boundary-spanning knowledge search and the cooperative innovation performance of non-core firms.

Given this background, this study aims to explore the following questions. Can boundary-spanning knowledge search contribute to the cooperative innovation performance of non-core firms in the context of sustainable development? What is the mechanism by which absorptive capacity impacts the cooperative innovation performance of these firms? For different types of boundary-spanning knowledge, are there differences in the mediating effect of absorptive capacity on cooperative innovation performance with respect to non-core firms in the context of sustainable development? To address these questions and explore the mechanism by which boundary-spanning knowledge search impacts the cooperative innovation performance of non-core firms in the context of sustainable developments, this study combines a resource-based view and organizational learning theory perspective to focus intensely on non-core firms as innovation agents with significant latecomer characteristics. The study then constructs a theoretical model that places absorptive capacity as a mediating variable, and the goal is to deepen theories related to boundary-spanning knowledge search in the context of cooperative innovation.

The rest of this paper is structured as follows: the “Literature Review and Research Hypothesis” section provides the theoretical background, formulates the hypotheses of this study, and constructs the theoretical framework. The “Research Design” section introduces the research methods. The “Analysis of Empirical Results” section presents the data analysis and results. The “Discussion” section discusses the conclusions, contributions, and limitations of this study.

Literature Review and Research Hypothesis

Boundary-Spanning Knowledge Search and Cooperative Innovation Performance of Non-Core Firms

Non-core firms refer to SMEs that are in a subordinate position in the innovation network compared to core firms (You & Wang, 2017). They tend to have the characteristics of latecomer enterprises, meaning they have a weak position with respect to market share, network position ranking, and technological innovation level (You & Wang, 2017). To make the definition of non-core firms more distinct, this paper refers to Li and Zhang (2020), who define non-core firms as SMEs at the end of the innovation network and which have innovation activities that are generally concentrated in the product production stage based on the innovation value chain structure.

Compared with core firms, non-core firms often face more constraints in the context of sustainable development, due to their smaller inferior size, status, and resource endowments (Khan et al., 2024). If leading firms break their development trajectories, rebuilding knowledge and progressing in the innovation value chain usually requires more investment and higher risks (Wang et al., 2017). Innovation firms in non-core positions lack the driving force for sustainable development, which is evident in their poor ability to search for and acquire resources in the innovation ecosystem. Additionally, the strategic sensitivity of non-core firms is low, making it difficult for them to effectively avoid risks (Kapoor & Agarwal, 2017; Zhu & Liu, 2018). When applying the concept of open innovation theory, when non-core firms are in an open innovation environment, external cooperation can make them more flexible and decisive when facing risks (Zhu et al., 2023). As such, establishing cooperative innovation alliances to identify, acquire, and allocate heterogeneous resources outside the firm's boundaries has gradually become an important paradigm for the innovation development of non-core firms (Janssen & Abbasiharoften, 2022).

Boundary-spanning knowledge search refers to the processes by which enterprises cross organizational knowledge boundaries, reform internal and external knowledge structures, and acquire heterogeneous resources through search behaviors (Leone et al., 2022; Nerkar & Miceli, 2016). When non-core firms participate in open cooperative innovation alliances, they may experience knowledge overload, because they face so much heterogeneous knowledge outside the enterprise boundary (Leone et al., 2022). Boundary-spanning knowledge search emphasizes active searching, which is a key way to help non-core firms make satisfactory decisions with limited resources (Lyu et al., 2022).

First, there is a clear goal-orientation and ability of sustainable development associated with knowledge search behavior, which can clarify the internal and external knowledge structure of firms and explore coupled heterogeneous knowledge in a more targeted way, by combining the real situation and actual needs of firms (Katila & Ahuja, 2017; Wang et al., 2023a, b). Second, boundary-spanning knowledge search can effectively mitigate knowledge overflow. The search behavior enriches the diversity of enterprise knowledge and can improve the existing knowledge system. Introducing novel heterogeneous knowledge can help expand the diversified innovation path of firms (Cunha et al., 2014).

This paper focuses on non-core firms in the context of sustainable development, which are a special subject in innovation studies and draws on Danneels's (2008) classification criteria with respect to knowledge types. The activity of boundary-spanning knowledge search is divided into two dimensions.

1. Boundary-spanning technological knowledge search and cooperative innovation performance of non-core firms

Boundary-spanning technological knowledge search includes enterprise behaviors that cross existing knowledge boundaries to identify and acquire knowledge resources (Lyu et al., 2022). Non-core firms acquire knowledge spillover by participating in cooperative alliances and consulting scientific research institutions. These effectively encourage the formation of complementarity between internal advantages and external heterogeneous resources, positively influencing firms' sustainable innovation behavior (Wang et al., 2023a, b).

The resource-based view posits that valuable, rare, imperfectly imitable, and non-substitutable resources (i.e., VRIN resources) play a key role in the sustainable development innovation of core technologies (Barney, 1991). In the context of poor environmental quality propositions, firms must examine the sustainability aspects of resources and technological innovation (Ahmad et al., 2023). Non-core firms usually find it challenging to generate VRIN resources on their own. Instead, they rely more on knowledge-seeking behaviors outside the firm's boundaries (Mei & Zhang, 2022; Zang et al., 2022). Boundary-spanning technological knowledge search is an important way for non-core firms to cross the boundary; they can flexibly adjust the search decision according to the cognitive structure of the enterprise and effectively improve the sharing efficiency of muted boundary-spanning knowledge.

Dynamic capability theory posits that absorbing effective resources outside the firm is critical for achieving innovation sustainable developed; innovation results from internal and external knowledge reconfiguration (Han et al., 2020; Leiponen & Helfat, 2010). However, the resource collection behavior of non-core firms may be passive based on the subjective willingness of core firms to disclose selectively (Rosenkopf & Nerkar, 2001). Boundary-spanning technological knowledge search helps non-core firms broaden the scope of knowledge search and break through the knowledge spillover boundaries created by selective disclosure by core firms, generating more high-value-added knowledge (Hao & Lu, 2022; Zhang & Fan, 2024).

Given factors such as resource endowments and risk preferences, non-core firms in the context of sustainable development often have a stronger tendency to experience path dependence in their technology development trajectory, and their technological iteration path may be influenced by the development of core firms (Garnsey & Leong, 2008; Mishrif & Khan, 2023). Boundary-spanning technological knowledge search can break the original technical barriers and circulate knowledge between the internal and external environment. This can help non-core firms avoid the disadvantage of being at the end of the network by gaining feedback faster, responding to the changes in the external market environment, and adjusting and changing the innovation strategy in a timely manner (Lopez-Vega et al., 2016; Wang et al., 2023a, b).

Based on the above discussion, this study proposes the following hypothesis.

H1: Boundary-spanning technological knowledge search has a positive effect on the cooperative innovation performance of non-core firms.

2. Boundary-spanning market knowledge search

Boundary-spanning market knowledge search reflects firms' spontaneous behavior to seek new markets and meet potential customers' needs externally (Lyu et al., 2022). Non-core firms achieve market and customer interactions through investigation and information collection, and bring timely feedback about market information back to the internal organization to guide enterprise sustainable innovation (Boulding & Christen 2003; Hiltebrand et al., 2015).

Organizational learning theory posits that the key to building a firm's innovation advantage and improving market opportunity identification efficiency lies in effectively combining a priori knowledge with dynamic market information (Politis, 2005). Sustainable innovative development requires non-core firms to have the ability to timely adjust their innovation strategies and innovation product performance to meet potential market demand. Boundary-spanning knowledge search can help firms broaden their horizons, strengthen their sensitivity to external market information, and dynamically tap the market strategy gap (Lifshitz-Assaf, 2018; Su et al., 2023a, b).

Expectation identity theory posits that customers' willingness to consume is determined by their degree of expectation and value perception of products or services. In other words, high customer expectations affect customer stickiness (Bhattacharjee, 2001). Specifically, consumers have pre-expectations with respect to products or services, and after their first consumption experience, they judge their satisfaction by comparing performance perceptions against expected value (Tarazona-Montoya et al., 2020). This degree of satisfaction determines customers' willingness to buy again, which expands the customer base, improve market share and further the innovation sustainable developed (Xiong et al., 2020).

Based on the above discussion, this study proposes the following hypothesis.

H2: Boundary-spanning market knowledge search has a positive effect on the cooperative innovation performance of non-core firms.

Absorptive Capacity and Cooperative Innovation Performance of Non-Core Firms

Absorptive capacity refers to the ability of firms to acquire, absorb, and transform external heterogeneous resources and then apply those resources (Cohen & Levinthal, 1990). There are four main functions associated with knowledge: acquiring knowledge, absorbing knowledge, transforming knowledge, and utilizing knowledge (Zahra & George, 2002). These can be simplified into two dimensions: potential absorptive capacity and actual absorptive capacity (Fosfuri & Tribo, 2008). Potential absorptive capacity mainly refers to the ability to acquire and absorb knowledge, which is a necessary precondition for the subject to carry out cooperative innovation (Fosfuri & Tribo, 2008). Actual absorptive capacity mainly includes the ability to transform and utilize knowledge, which is needed for the subject to maintain a competitive advantage with respect to cooperative innovation (Fosfuri & Tribo, 2008).

Potential absorptive capacity can effectively screen the value of external knowledge spillovers. Cooperative innovation alliances usually search for and transfer as much complementary knowledge as possible to generate actual outcome benefits. However, firms do not fully benefit from knowledge-intensive innovation, and even high-tech firms face the dilemma of knowledge redundancy (Escrineo et al., 2009). Potential absorptive capacity is the ability of non-core firms to acquire and absorb heterogeneous resources by deconstructing the value of new knowledge and adapting external knowledge to existing knowledge structures (Todorova & Durisin, 2007). Given constraints in technological innovation capability and innovation resource endowments, non-core firms in sustainable development may find that the existing technology chain cannot absorb and match the heterogeneous knowledge when deconstructing external knowledge, coding it, and reorganizing it to fit into the internal knowledge system. This may lead non-core firms to prioritize familiar and easy-to-handle knowledge and not consider new knowledge with potential value (Skippari et al., 2017). This can lead to the cart before the horse in boundary-spanning knowledge search, wasting innovation costs. Potential absorptive capacity can help non-core firms in the context of sustainable development to analyze the advantages and disadvantages of external knowledge and identify the value of knowledge overflowing from the external environment, such as from cooperative firms and research institutions (Mim et al., 2022). This can help broaden the knowledge architecture and mitigate the shortcomings of a lack of knowledge stock and weak innovation technology (Kostopoulos et al., 2011).

In contrast, a firm's actual absorptive capacity can accelerate the circulation of innovative knowledge and the realization of its value. The process of transferring and grounding knowledge resources can be difficult, especially for non-core firms with significant latecomer characteristics. As such, some initiatives to handle innovative knowledge may not positively impact results (Yildiz et al., 2024). Actual absorptive capacity is the firm's ability to transform and apply complementary resources; this means there is a direct connection with the value of the commercial application of knowledge resources in the market environment for sustainable development. This occurs by dismantling the theoretical basis of hetero-technical knowledge to improve the effectiveness of innovative activities (Gonzalez, 2024). The value and industrial chains of non-core firms in the context of sustainable development are relatively underdeveloped, and their sensitivity to new technologies is relatively slow. This makes it difficult for them

to effectively handle new knowledge acquired through potential absorptive capacity and can make it difficult for them to benefit from it (Khan et al., 2019). Actual absorptive capacity, however, can deepen the firms' ability to transform and apply knowledge resources, improve the ability of subjects to circulate innovation with respect to external coupled resources and encourage innovation results. This benefits cooperative innovation (Mu & Jiang, 2024).

Based on the above discussion, this study proposes the following hypothesis.

H3a: Potential absorptive capacity has a positive effect on cooperative innovation performance.

H3b: Actual absorptive capacity has a positive effect on cooperative innovation performance.

The Mediating Role of Absorptive Capacity

Boundary-spanning technological knowledge search can identify complementary knowledge related to the internal systems of non-core firms through potential absorptive capacity. Boundary-spanning technological knowledge search can capture a large amount of differential technological knowledge in the external environment for innovation agents. However, this knowledge is often disorderly before it is applied to innovation activities. Potential absorptive capacity helps the subject to conduct the initial organization of this information by implementing the identification, coding, and reconstruction of disordered knowledge. It also lays the foundation for systematically internalizing the knowledge into cooperative innovation results (Gonzalez, 2024). However, using potential absorptive capacity, boundary-spanning market knowledge search can quickly break through local constraints. Boundary-spanning market knowledge search captures cross-domain market knowledge for innovation subjects to improve their internal knowledge architecture. However, there remains the possibility of conflict between heterogeneous knowledge systems and non-core firms' technological systems. Through potential absorptive capacity, non-core firms in the context of sustainable development will anticipate the external environment when conducting boundary-spanning knowledge searches and adjust the scale of action according to their own needs to achieve high-quality search (Collins & Smith, 2006). Moreover, strong potential absorptive capacity strengthens non-core firms' responsiveness to customer needs, increases the integration of new and different domain knowledge with existing knowledge systems, and avoids the core firm's rigidity trap. This helps improve their cooperative innovation performance (Kostopoulos et al., 2011).

Based on the above discussion, this study proposes the following hypothesis.

H4a: Potential absorptive capacity mediates the relationship between boundary-spanning technological knowledge searches and non-core firms' cooperative innovation performance.

H4b: Potential absorptive capacity mediates the relationship between boundary-spanning market knowledge searches and non-core firms' cooperative innovation performance.

Boundary-spanning technological knowledge search can contribute to the realization of complementary knowledge values through actual absorptive capacity. A strong actual absorptive capacity implies there is a relatively low conversion cost with respect to knowledge resources. This supports the expansion of the knowledge stock of non-core firms, optimization of the core production chain, and the renewal and modernization of core technologies (Lyu et al., 2022). Actual absorptive capacity transfers the high value-added knowledge acquired through boundary-spanning knowledge search through such mechanisms as the technology and innovation chains. The output and the internalized knowledge are used to optimize existing products or design new products, and the results are reflected in the form of cooperative innovation performance (Jansen et al., 2005).

Boundary-spanning market knowledge search can generate competitive advantages in the market through actual absorptive capacity. Actual absorptive capacity enables knowledge assimilation and can help the firm accept and integrate the sustainable development market knowledge and potential customer knowledge (Fan et al., 2023). Non-core firms in the context of sustainable development explore differentiated fields using boundary-spanning knowledge search activities and acquire novel market knowledge fragments from upstream and downstream industry firms. These fragments are integrated with the existing knowledge system of the innovation subject through knowledge assimilation and the application capability associated with actual absorptive capability. This can catalyze the generation of new insights (Katila & Chen, 2008). These new ideas can guide the sustainable development of breakthrough cooperative innovation, enhancing the inimitable value of cooperative innovation products.

Based on the above discussion, the study proposes the following hypothesis.

H5a: Actual absorptive capacity mediates the relationship between boundary-spanning technological knowledge search and non-core firms' cooperative innovation performance.

H5b: Actual absorptive capacity mediates the relationship between boundary-spanning market knowledge search and non-core firms' cooperative innovation performance.

Based on the analysis above, Fig. 1 provides a relationship model for this study.

Research Design

Sample and Data Collection

For analyzing the mechanism between boundary-spanning knowledge search, absorptive ability, and cooperative innovation performance of non-core firms, as

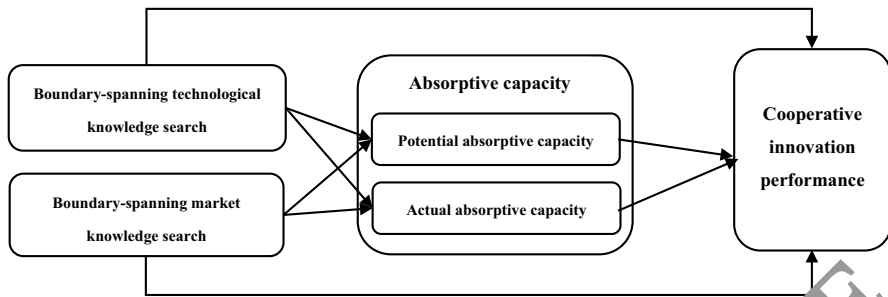


Fig. 1 The theoretical framework

well as to take into account the feasibility and convenience of data collection, the sample of non-core firms in Yunnan Province, where the research team is based, is chosen. In order to define the specific group of non-core firms, the cooperative patent data of Yunnan enterprises from 2015 to 2019 was searched (the reason for excluding the cooperative patent data of 2020 and 2021 is that the approval status is unknown), and finally, 1,273 cooperative patent data were retrieved. We denote enterprises ranked after 70% of the annual network position as non-core enterprises. All eligible non-core firms within the scope of the studied year were screened out, and by deleting duplicate firms, 342 non-core firms were ultimately obtained. Again, with the assistance of the Yunnan Provincial Department of Science and Technology and Yunnan Small and Medium Enterprise Service Centre, we gathered the relevant contact information of the senior managers of the aforementioned non-core firms and contacted each one individually to determine their willingness. A total of 270 non-core firms whose senior managers agreed to take part in the survey were finally identified as the sample firms, and a roster was made. The descriptive statistical information about the sample firms is presented in Table 1.

Due to the wide application and universality of the questionnaire method in the field of management, the study used it for data collection and hypothesis testing. With the exception of control variables, the variables were measured on a 7-point Likert scale (where 1 indicates completely disagree and 7 indicates completely agree). In the process of sample collection, to ensure the accuracy and rationality of the formal survey, we conducted a small-scale pilot study as preparatory work. The pilot survey was distributed to two main groups. On the one hand, it was distributed to students in the MBA classes of the research team's school who met the selection criteria: (1) Employed by the sample companies; (2) Holding positions of senior manager or above. A total of 27 individuals were selected, of whom 20 agreed to participate in the pilot study. On the other hand, 50 sample companies were randomly selected, and questionnaires were distributed via email. A total of 70 questionnaires were distributed in the pilot study, and 51 valid questionnaires were recovered, resulting in an effective recovery rate of 72.85%. This process lasted for 2 months.

We analyzed the recovered questionnaires, refined and optimized the items, and formed the final version of the survey questionnaire. Subsequently, we conducted the formal questionnaire survey, and sent to the top managers of the sample companies

Table 1 Characteristics of samples

Characteristic	Classification	Frequency	Percentage (%)
Years	<3 years	64	35.2
	3–10 years	95	52.2
	>10 years	23	12.6
Nature	State-owned enterprises	21	11.5
	Private enterprises	121	66.5
	Others	40	22
Industry	The electronic/communication equipment industry	13	7.1
	The biomedical/chemical industry	15	8.2
	The new energy industry	15	8.2
	The software /IT service industry	25	13.7
	The electrical/mechanical manufacturing industry	22	12.1
	Others	97	50.5
Size	Small	118	64.8
	Medium	64	35.2

by email and offline delivery. The questionnaire content includes an introduction (briefly explaining the purpose and content of the survey to the respondents, and informing them that there are no right or wrong answers to any of the items, to ensure that respondents fill out the questionnaire truthfully), the main body of the questionnaire (items related to cross-boundary knowledge search, absorptive capacity, and the innovation performance of non-core enterprise cooperation), basic information about the respondents and their companies, and a closing statement (expressing gratitude to the respondents and inquiring about any unreasonable aspects of the questionnaire design). Among the 270 questionnaires distributed, 198 were returned, representing a return rate of 73.3%. After eliminating invalid questionnaires with blank or missing answers, a total of 182 valid questionnaires were returned, with an effective rate of 67.4%. This phase lasted for 3 months. Throughout all process, we strictly adhered to international and national standards and briefly introduced measures to ensure data confidentiality to participants.

Measurement of Variables

Independent Variables

As was previously mentioned, boundary-spanning knowledge search is utilized as an independent variable in this paper in order to explore the direct effects of its different dimensions—specifically, the two dimensions of boundary-spanning technological knowledge search and boundary-spanning market knowledge search—on cooperative innovation performance. The scale used for boundary-spanning technological knowledge search (denoted as *Tkb*) is based primarily on the research findings of Sidhu et al. (2007) and Xiong et al. (2020) and is measured by setting six questions,

such as “emphasis on the development of industries related to the company’s technology,” etc. The scale employed for searching boundary-spanning market knowledge (denoted as *Mkb*) is mainly based on the research results of Granovetter (2010) and is measured by setting 5 items, which includes “good at collecting information on the preferences of potential customers.”

Dependent Variables

As mentioned earlier, this paper uses cooperative innovation performance as the dependent variable. For the purpose of measuring the cooperative innovation performance (denoted as *Cip*), technological innovation performance is used as a direct test of cooperative innovation performance by referring to the research results of Hagedoorn and Duysters (2002), and is determined by setting five questions, such as “Through cooperation, new products are developed faster than competitors.” In addition, referring to Flynn et al. (2010), management innovation performance was adopted as a robustness test and measured by setting four items, including “through cooperation, strategic planning goals are more clearly defined,” etc.

Mediating Variable

Absorptive capacity is introduced experimentally so as to investigate the mediating impact of its various dimensions (potential absorptive capacity and actual absorptive capacity) on the relationship between the explanatory and the explained variables. Following the studies of Lichtenthaler and Lichtenthaler (2009), the potential absorptive capacity (denoted as *Pac*) is measured by developing six questions, such as “Your company is able to recognize changes in the market,” and the actual absorptive capacity (denoted as *Aac*) is measured by setting up six questions, such as “Your company is able to recognize changes in the market Table 2.”

Control Variables

In this paper, years (denoted by *Years*), the nature of the enterprise (denoted by *Nature*), the industry (denoted by *Industry*), and the enterprise size (denoted by *Size*) are selected as control variables to account for differences among enterprises.

The reliability of the variables involved in this study was tested using SPSS 23.0, and the results of the analysis are illustrated in Table 1. The Cronbach’s α coefficients of boundary-spanning technological knowledge search, boundary-spanning market knowledge search, cooperative innovation performance, potential absorptive capacity, and actual absorptive capacity are all greater than 0.7, demonstrating that the scale has good reliability; the KMO values of boundary-spanning knowledge search, cooperative innovation performance, and absorptive capacity are all significantly higher than 0.7, and the scale has high structural validity, and the sample data is suitable for factor analysis. The standardized loading coefficients of all question

Table 2 Reliability and validity test of the constructs

Construct	KMO	Dimension	Factor loading	Cronach's α	CR	AVE
Boundary-spanning knowledge search	0.943	<i>Kkb</i>	0.730	0.917	0.882	0.556
			0.837			
			0.768			
			0.796			
			0.682			
		<i>Mkb</i>	0.443	0.903	0.864	0.562
			0.86			
			0.704			
			0.819			
			0.670			
Cooperative innovation performance	0.915	<i>Cip</i>	0.668	0.917	0.875	0.584
			0.718			
			0.753			
			0.739			
			0.755			
		<i>Cips</i>	0.854	0.926	0.888	0.665
			0.795			
			0.806			
			0.887			
			0.771			

Table 2 (continued)

Construct	KMO	Dimension	Factor loading	Cronbach's α	CR	AVE
Absorptive capacity	0.935	Ac	0.836	0.903	0.873	0.634
			0.764			
			0.830			
			0.750			
Ac			0.847	0.960	0.921	0.656
			0.847			
			0.780			
			0.833			
			0.813			
			0.784			

Table 3 Correlation analyses of the variables

Items	Years	Nature	Industry	Size	Tkb	Mkb	Pac	Aac	Cip
Years	-								
Nature	-0.217**	-							
Industry	0.073	0.100	-						
Size	-0.028	0.047	0.084	-					
Tkb	-0.043	-0.115	-0.032	-0.067	0.746				
Mkb	-0.101	0.080	-0.097	-0.049	0.000	0.750			
Pac	0.034	-0.109	-0.067	-0.006	0.393***	0.546***	0.796		
Aac	-0.118	0.036	-0.007	0.018	0.441***	0.300***	0.000	0.810	
Cip	-0.150*	0.035	-0.099	0.082	0.353***	0.391***	0.262***	0.495***	0.764

*Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted

items are greater than 0.6; the average extracted variance (AVE) is greater than 0.5, and the combined reliability (CR) values are above 0.7, proving the good convergent validity of the scale. Meanwhile, as depicted in Table 3, the square root of AVE for each dimension given by the diagonal line is in the range of [0.746, 0.810], which is significantly higher in comparison to the maximum value of the correlation coefficient of each factor, 0.546, proving that the scale has good discriminant validity.

Reliability Test

The correlation coefficients between the variables are indicated in Table 3. The correlation coefficients of each variable are within the acceptable range, and there are significant positive correlations between the two variables of boundary-spanning search, absorptive capacity, and cooperative innovation performance, and the hypotheses can be initially verified. The correlation coefficients between the variables are consistent with expectations, which provide preliminary support for the subsequent hypothesis testing analysis.

Analysis of Empirical Results

In the context of sustainable development, boundary-spanning knowledge search is beneficial to cooperative innovation. Based on the data of non-core firms of Yunnan in 2015 to 2019, this paper reveals that boundary-spanning technological knowledge search and boundary-spanning market knowledge search are beneficial to cooperative innovation performance. Moreover, absorptive capacity mediates the relationship between boundary-spanning technological knowledge searches and non-core firms' cooperative innovation performance and actual absorptive capacity mediates the relationship between boundary-spanning market knowledge search and non-core firms' cooperative innovation performance. Yet, the mediating effect of potential absorptive

Table 4 Tests of main effect

Control variables	Dependent variables Cip				
	Model ₁	Model ₂	Model ₃	Model ₄	Model ₅
Years	−0.211	−0.172	−0.164	−0.217	−0.126
Nature	0.020	0.099	−0.030	0.066	0.011
Industry	−0.059	−0.057	−0.037	−0.049	−0.059
Size	0.178*	0.225	0.216	0.176	0.164
Independent variables					
Tkb		0.359***			
Mkb			0.380***		
Pac				0.266***	
Aac					0.482***
R ²	0.038	0.164	0.179	0.177	0.267
Adjusted R ²	0.016	0.140	0.155	0.152	0.246
F-value	1.737*	6.884***	7.656***	4.229**	12.833***
Durbin-Watson	1.810	1.902	1.765	1.786	1.835

*Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted

capacity on the relationship between boundary-spanning market knowledge search and non-core firms' cooperative innovation performance is invalid. Research results indicate firms should pay attention to the influence of boundary-spanning knowledge search on cooperative innovation and do their best to promote cooperative innovation performance in this sustainable development era.

Main Effect

Using SPSS 23.0, the major impacts were investigated in order to build a model of the relationship between the variables. As shown in Table 4, the unmediated variable model was first constructed. Moreover, the control variables, such as the year of establishment, the nature of the enterprise, the industry to which it belongs, and the scale of the enterprise, were introduced as the independent variables, and the cooperative innovation performance was utilized as the dependent variable to construct Model 1. Subsequently, the independent variables boundary-spanning technological knowledge search, boundary-spanning market knowledge search, potential absorptive capacity, and actual absorptive capacity were added based on model 1 for constructing models 2 to 5, respectively. The regression results in Table 4 point out that boundary-spanning technological knowledge search, boundary-spanning market knowledge search, potential absorptive capacity, and actual absorptive capacity all have a positive impact on cooperative innovation performance (β values are 0.359, 0.380, 0.266, and 0.482, respectively, with p values less than 0.001), and H1, H2, H3a, and H3b are all verified. To understand the relationship between the independent and dependent variables, the results are graphically illustrated in Fig. 2.

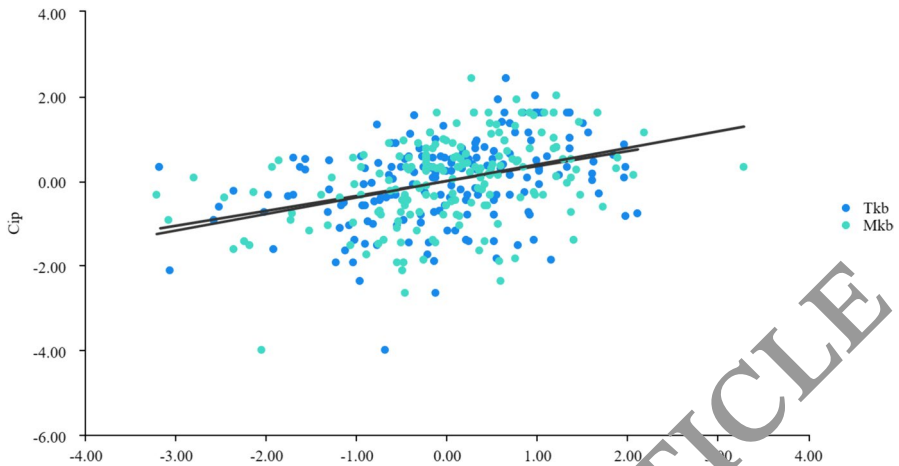


Fig. 2 Main effect

Mediation Effect

The mediation effect of potential absorptive capacity and actual absorptive capacity was verified by drawing on Ma et al.'s (2022) mediation effect verification practice (2022), with specific results in Table 5. As can be seen from Table 5, model 6 was constructed by introducing boundary-spanning technical knowledge search as the independent variable and potential absorptive capacity as the dependent variable; subsequently, model 7 was constructed by adding potential absorptive capacity based on model 2; model 8 was built by using boundary-spanning market knowledge search as the independent variable and potential absorptive capacity as the dependent variable, meanwhile, model 9 was constructed by incorporating potential absorptive capacity based on model 3.

It is clear from the regression findings of model 6 that the potential absorptive capacity is positively and significantly impacted by boundary-spanning technological knowledge search ($\beta=0.389$, $p<0.001$); comparing the regression results of model 2 and model 7, it can be observed that when the mediating variable of potential absorptive capacity is added, boundary-spanning technological knowledge search still has a considerable effect on cooperative innovation performance but its coefficient of the effect decreases (β value decreases from 0.359 to 0.301). The regression coefficient of potential absorptive capacity is also significant ($\beta=0.149$, $p<0.05$). This indicates that potential absorptive capacity partially mediates the relationship between boundary-spanning technological knowledge search and cooperative innovation performance, and H4a holds.

The regression results of model 8 show that boundary-spanning market knowledge search has a positive and significant effect on potential absorptive capacity ($\beta=0.565$, $p<0.001$). When comparing the regression results of model 3 and model 9, it can be seen that when the mediating variable of potential absorptive capacity is added, even though boundary-spanning market knowledge capacity still has a

Table 5 Test of mediation effect for absorptive capacity

Control variables	Pac Model ₆	Cip Model ₇	Pac Model ₈	Cip Model ₉	Aac Model ₁₀	Cip Model ₁₁	Aac Model ₁₂	Cip Model ₁₃
Years	0.067	−0.182	0.094	−0.172	−0.127	−0.120	−0.140	−0.108
Nature	−0.089	0.112	−0.248*	−0.010	0.117	0.052	−0.020	−0.021
Industry	−0.033	−0.052	−0.003	−0.037	0.002	−0.058	0.017	−0.044
Size	0.059	0.216	0.064	0.211	0.088	0.190	0.058	0.192
Independent variables								
Tkb	0.389***	0.301***			0.448***	0.178*		
Mkb			0.565***	0.336***			0.296***	0.250***
Pac		0.149*		0.078				
Aac						0.403***		0.406***
R ²	0.164	0.182	0.326	0.183	0.211	0.292	0.100	0.327
Adjusted R ²	0.140	0.154	0.307	0.155	0.189	0.268	0.074	0.304
F-value	6.885***	6.501***	17.047***	6.520***	9.411***	12.025***	3.891*	14.194***
Durbin-Watson	2.031	1.878	1.992	1.765	1.690	1.871	1.739	1.754

*Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted

significant and positive influence on cooperative innovation performance (β value reduces from 0.380 to 0.336, p values are less than 0.001), potential absorptive capacity does not show a significant effect on cooperative innovation performance ($\beta = 0.078$, $p > 0.05$). This suggests that potential absorptive capacity did not significantly mediate the association between the boundary-spanning market knowledge search and cooperative innovation performance, and H4b was not verified.

As Table 5 highlights, model 10 was constructed by introducing boundary-spanning technological knowledge search as the independent variable and actual absorptive capacity as the dependent variable; subsequently, model 11 was developed by adding actual absorptive capacity to model 2; model 12 was constructed utilizing boundary-spanning market knowledge search as the independent variable and actual absorptive capacity as the dependent variable; meanwhile, model 12 was constructed by introducing actual absorptive capacity to model 3. Model 13 was constructed.

It can be observed from the regression results of model 10 that boundary-spanning technological knowledge search has a positive as well as significant effect on actual absorptive capacity ($\beta = 0.448$, $p < 0.001$); by comparing the regression results of model 2 and model 11, it can be found that when the mediating variable of actual absorptive capacity is added, the coefficient of the effect of boundary-spanning technological knowledge search on cooperative innovation performance remains significant. However, its coefficient of effect reduces (β value from 0.359 to 0.178), and the regression coefficient of actual absorptive capacity itself is also significant ($\beta = 0.403$, $p < 0.001$). This indicates that the relationship between boundary-spanning technological knowledge search and cooperative innovation performance is partially mediated by actual absorptive capacity, as verified by H5a.

The regression results from model 12 show that boundary-spanning market knowledge search has a positive and significant impact on actual absorptive

capacity ($\beta = 0.296$, $p < 0.001$); the regression results from model 3 and model 13 illustrate that when the mediating variable of actual absorptive capacity is added, the coefficient of the effect of boundary-spanning market knowledge search on cooperative innovation performance remains significant however, its effect coefficient decreases (β value from 0.380 to 0.260), plus the regression coefficient of actual absorptive capacity itself is also significant ($\beta = 0.406$, $p < 0.001$). This highlights that actual absorptive capacity partially mediates the association between boundary-spanning market knowledge search and cooperative innovation performance, as verified by H5b.

Possible explanations for why potential absorptive capacity did not act as a mediating factor in the relationship between the boundary-spanning market knowledge search and cooperative innovation performance (i.e., H4b is not verified) include the following: (1) the relatively weak learning intentions of non-core firms may have difficulty in activating the performance of their potential absorptive capacity mediating path. Non-core firms may be more likely to pursue short-term vested profits or catch up with market trends guided by core firms because they are constrained by the relative lack of knowledge stock, relatively homogeneous organizational structure, relatively weak innovation technology, and controlled by core firms (Wang et al., 2017). They also tend to prioritize the absorption of shallow market information that matches their knowledge systems, thereby inhibiting their intention to enhance learning through boundary-spanning search to acquire disruptive knowledge. The possibility may limit the mediating effect of its potential absorptive capacity to a certain extent (Wang et al., 2020). (2) there may be a non-fit relationship between potential absorptive capacity and boundary-spanning market knowledge search. As mentioned earlier, the desire of non-core firms to pursue technological catch-up is more urgent, which may lead them to invest huge boundary-spanning search costs for the purpose of breaking the status constraint. However, because the market environment is time-varying, market knowledge is more complicated and subjectively motivated than technical knowledge that can be acquired by updating technical systems, making the boundary-spanning market knowledge search more challenging and obscure (Lu & Chen, 2021). The relatively limited absorptive capacity of non-core firms may make it challenging to activate their intermediary mechanisms, resulting in low efficiency in the transformation of boundary-spanning search results, which in turn leads to wasted innovation costs as well as poor innovation performance.

Robustness Tests

In order to verify the robustness of the previous empirical results, the model was regressed again by replacing technological innovation performance with management innovation performance as the dependent variable with reference to the variable replacement method in the literature (Zheng, 2021), and the final results of the robustness tests were obtained, which are presented in Tables 6 and

Table 6 Robustness tests of main effect

Control variables	Cip				
	Model ₁₄	Model ₁₅	Model ₁₆	Model ₁₇	Model ₁₈
Years	0.012	0.047	0.052	0.002*	0.069
Nature	−0.041	0.030	−0.084	0.031	−0.047
Industry	−0.042	−0.040	−0.023	−0.027	−0.042
Size	−0.283*	−0.240	−0.251	−0.286	−0.292*
Independent variables					
Tkb		0.323***			
Mkb			0.326***		
Pac				0.416***	
Aac					0.324***
R^2	0.026	0.128	0.130	0.096	0.130
Adjusted R^2	0.004	0.103	0.105	0.175	0.105
F -value	1.184*	5.176***	5.237***	5.94***	5.243***
Durbin-Watson	1.859	1.838	2.016	1.710	1.973

*Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted

7. The regression results demonstrate that, despite the substantial size changes in the regression coefficients of the models, their signs and significance results remained consistent with the previous regression results. This indicates that the findings of this paper have some robustness.

Table 7 Robustness tests of mediation effect

Control variables	Pac Model ₁₉	Cip Model ₂₀	Pac Model ₂₁	Cip Model ₂₂	Aac Model ₂₃	Cip Model ₂₄	Aac Model ₂₅	Cip Model ₂₆
Years	0.067	0.024	0.094	0.020	−0.127	0.076	−0.140	0.087
Nature	−0.029	0.061	−0.248*	0.001	0.117	0.004	−0.020	−0.078
Industry	−0.033	−0.029	−0.003	−0.022	0.002	−0.041	0.017	−0.027
Size	0.059	−0.261	0.064	−0.272	0.088	−0.260	0.058	−0.265
Independent variables								
Tkb	0.389***	0.190*			0.448***	0.222*		
Mkb			0.565***	0.133*			0.296***	0.252**
Pac		0.342***		0.341				
Aac						0.225*		0.251**
R^2	0.164	0.226	0.326	0.208	0.211	0.168	0.100	0.186
Adjusted R^2	0.140	0.200	0.307	0.181	0.189	0.140	0.074	0.158
F -value	6.885***	8.526***	17.047***	7.661***	9.411***	5.901***	3.891**	6.666***
Durbin-Watson	2.031	1.722	1.992	1.810	1.690	1.923	1.739	2.062

*Significant at $p < 0.1$; **Significant at $p < 0.05$; ***Significant at $p < 0.001$. The constant terms are omitted

Discussion

Based on organizational search theory, technology catch-up theory, and open innovation theory, the study constructs a model of the relationship between boundary-spanning search and cooperative innovation performance of non-core firms in the context of sustainable development while introducing absorptive capacity as a mediating variable so as to explore the mediating role of potential absorptive capacity and actual absorptive capacity on the relationship between boundary-spanning search and cooperative innovation performance of non-core firms in the context of sustainable development, respectively.

The cooperative innovation performance of non-core firms in the context of sustainable development is positively impacted by boundary-spanning search. The regression findings suggest that regardless of whether the dependent variable is technological innovation performance or managerial innovation performance, the boundary-spanning technological knowledge search and the boundary-spanning market knowledge search show a considerable positive effect on the cooperative innovation performance of non-core firms. This helps demonstrate the value of boundary-spanning knowledge search for non-core firms to achieve innovation sustainable developed.

Absorptive capacity has a dual mechanism of action on the cooperative innovation performance of non-core firms in the context of sustainable development. The empirical results signify that, on the one hand, both potential and actual absorptive capacity have a significant positive direct effect on the cooperative innovation performance of non-core firms; on the other hand, absorptive capacity also plays a mediating role in the correlation among boundary-spanning knowledge search and the cooperative innovation performance of non-core firms, i.e., the boundary-spanning knowledge search can indirectly influence the cooperative sustainable innovation performance of non-core firms via the mediating mechanism of absorptive capacity.

Absorptive capacity dimensions mediate the relationship between boundary-spanning knowledge search and cooperative innovation performance of non-core firms. In particular, the relationship between boundary-spanning knowledge search (whether it is boundary-spanning technological knowledge search or boundary-spanning market knowledge search) and non-core firms' cooperative innovation performance is partially mediated by actual absorptive capacity. In contrast, the relationship between boundary-spanning technological knowledge search and cooperative innovation performance is only partially mediated by potential absorptive capacity and does not significantly mediate the relationship between boundary-spanning market knowledge search and cooperative innovation performance.

Theoretical Contributions

Firstly, in contrast to earlier literature on cooperative innovation that takes pan-type enterprises as the research object, this paper places a great emphasis on the cooperative innovation performance of non-core firms in the context of sustainable development as a specific research object and attempts to analyze its antecedent mechanism

from the standpoint of boundary-spanning knowledge search and absorptive capacity, thus deepening the perspective and depth of research on cooperative innovation of non-core firms.

Second, differing from the previous research literature, this paper draws on the research results of scholars such as Sidhu et al. (2007) and Granovetter (2010), to experimentally utilize boundary-spanning knowledge search as an antecedent variable (rather than a mediating variable) and investigate the direct and indirect mechanisms of its various dimensions on the cooperative innovation performance of non-core firms, so as to thoroughly explore the multiple explanatory paths of boundary-spanning knowledge search on cooperative innovation performance.

Third, differing from the prior research literature that habitually treats absorptive capacity as an independent or moderating variable, this paper incorporates it as a mediating variable in the model of the relationship between boundary-spanning knowledge search and non-core firms' cooperative innovation performance by combining the studies of Zahra and George (2002), Lichtenthaler and Lichtenthaler (2009), etc. The empirical results finally revealed the heterogeneous mediating role of its different dimensions on the aforementioned relationship, effectively expanding the entry perspective and the function of absorptive capacity in interpreting the variability of cooperative innovation performance.

Managerial Implications

First, we concentrate on guiding non-core firms in the context of sustainable development to carry out boundary-spanning knowledge search activities. The empirical findings reveal that all dimensions of boundary-spanning knowledge search show a significant positive effect on the cooperative innovation performance of non-core firms. This indicates that relying on boundary-spanning knowledge search behavior is beneficial for non-core firms to achieve technological catch-up and innovation breakthroughs. As a result, non-core firms should be encouraged to actively engage in boundary-spanning knowledge search activities in order to continuously diversify their heterogeneous technological and market knowledge. Specifically, on the one hand, non-core firms should strengthen their adaptability to the changing core technology paradigm, break the time and space constraints of firms by forming technological innovation alliances, and actively promote boundary-spanning knowledge search activities of technological knowledge to efficiently capture and allocate resources required for innovation and developing new channels and paths for cooperative innovation; on the other hand, non-core firms should have the courage to break the boundaries of markets and industries and establish direct communication channels with markets and customers to strengthen their excavation of market information and customers' real needs.

Secondly, the absorptive capacity of non-core firms in sustainable development is actively undertaken to undertake the mechanism. According to the empirical results, absorptive capacity is known to show two mechanisms of direct and indirect intermediation on the cooperative innovation performance of non-core firms. For this reason, non-core firms should take the initiative to develop the absorptive capacity-inducing mechanism while planning to improve absorptive capacity in the overall strategic

deployment. To be more specific, on the one hand, by establishing a sound employee recruitment and training system, as well as by fostering a good learning atmosphere and a high-quality learning organization, non-core firms can have employee resources with strong professional knowledge and business ability it is possible to ensure the identification and acquisition of technological and market knowledge by non-core firms. On the other hand, by strengthening the construction efforts of R&D investment, an information platform with sharing and immediacy can be completed, which promotes the integration and circulation of new knowledge and old knowledge and enhances the utilization and digestion ability of external overflow resources.

Conclusion

The purpose of this paper is to explore the influence of boundary-spanning technological and market knowledge search on tacit cooperative innovation by exploring the mediating role of absorptive capacity among non-core firms in the context of sustainable development. Drawing on the knowledge-based view, this study constructs a theoretical framework to explore the effects of boundary-spanning knowledge search on cooperative innovation. Going further, because of the specific traits of boundary-spanning technological and market knowledge search, they play different roles during the cooperative innovation process in non-firms.

This study also emphasizes the innovative behavior of non-core firms in the context of sustainable development. Overall, knowledge resources can impact the sustainable innovative development of non-core firms. Boundary-spanning knowledge search provides a rich resource reservoir for cooperative innovation of non-core firms, and absorptive capacity catalyzes the transformation and implementation of heterogeneous knowledge. Both factors and their interconnection can ensure the improvement of cooperative innovation performance of non-core firms under sustainable development.

Limitations and Future Research

The study also has the following limitations: First, considering the reliability of the sample source as well as the feasibility of data collection, the sample data in this paper primarily comes from a single region, Yunnan Province, which may prevent the conclusions drawn in this paper from being generalizable. For this reason, the subsequent study can expand the scope of the selection of the tested enterprises and assess the applicability of the empirical findings in other regions. Second, the study mostly relies on static cross-sectional data in order to verify the relationship between boundary-spanning knowledge search, absorptive capacity, and cooperative innovation performance, which may neglect the time-varying and robustness of the findings. To address this potential issue, we will incorporate time series into the sample selection criteria in subsequent research.

Funding The paper was supported by the National Natural Science Foundation Project of China (Grant Number: 72264039), the Foundation Project of Yunnan Province's Ten Thousand Talents Plan for Top Young Talents, the Yunnan Fundamental Research Projects (Grant Number: 202401AT070282) and the Scientific Research Project of the Department of Education of Yunnan Province (Grant Number: 2023Y0669). The authors thank them sincerely for supporting the paper's funding.

Data Availability Data will be made available on request.

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